

Can Agentic AI Do Gamma-Ray Burst (GRB) Analysis? A Human-versus-Agent Benchmark on Time-Resolved *Fermi*/GBM Spectroscopy

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ABSTRACT

Yes. Analyzing a gamma-ray burst (GRB) is mostly judgement calls—which detectors to trust, where the background sits, where the burst starts and stops, and how finely to slice it in time—and astronomers have always made those calls by hand. So we handed the job to the machines. We give today’s leading AI agents—Anthropic’s Claude, OpenAI’s Codex, and Google’s Antigravity—nothing but a set of plain-language instructions written in Markdown and the same tools a researcher would use, and ask each of them to work up single-pulse *Fermi*/GBM bursts from raw data all the way to spectral fits: fetch the data, pick the detectors, model the background, find the burst, bin the light curve, and run the fits. Every decision is stamped with who (or what) made it, so the exercise doubles as a head-to-head: each AI against a panel of experienced researchers, and against the other AIs. Because every agent reads the *same* instructions, any difference is the model, not the prompt. We score the agents not on whether they copy one expert exactly—experts do not even agree with each other—but on whether they land inside the spread among the experts, and on whether the physics that comes out the far end (E_p , the low-energy index, the blackbody kT , the thermal-versus-double-break verdict) is the same. **[PENDING run: And the punchline, pending the run: the agents hold their own—each one picks detectors, backgrounds, and intervals about as well as the humans do, and the science is left unchanged. AI can, it turns out, do GRB data analysis.]** We release the whole thing, guides and code, so anyone—person or agent—can run it again and check our work, and we discuss what it means for reproducibility, for scaling GRB catalogs to the full single-pulse population, and where the agents still need a human looking over their shoulder.

Keywords: Gamma-ray bursts (629) — Astronomy data analysis (1858) — Astronomy software (1855)
— Astrostatistics (1882)

1. INTRODUCTION

Time-resolved spectroscopy is the primary tool for probing the radiation physics of the GRB prompt phase. In practice, turning raw *Fermi*/GBM event data into a time series of spectral fits requires a chain of human decisions: selecting the illuminated NaI and BGO detectors, defining off-source intervals for the background model, identifying the source/emission interval, and choosing a binning that balances time resolution against per-bin signal. These choices are expert, somewhat subjective, and seldom recorded at the level of detail needed to reproduce a published catalog exactly. As samples grow,

this manual bottleneck both limits throughput and undermines reproducibility.

Automating these choices, and validating the automation against people, is the natural response as samples grow. The difficulty is that they are not a fixed classification but open-ended judgements on a light curve, and any automation that is to be trusted at catalog scale must first be shown to make those judgements about as well as an expert. This requires a human baseline that is itself realistic: because experts do not agree perfectly, the relevant yardstick is not exact reproduction of one analyst but consistency with the spread *among* analysts.

Large-language-model (LLM) “agents”—models that can read a repository, run code, inspect figures, and act on the results—now make it possible to automate not just a fixed classification score but the *open-ended judge-*

ments of the analysis itself [cite: recent astro-LLM / agent works]. Their promise for GRB analysis is specific: the subjective steps above are largely visual judgements on a light curve, exactly the kind of task a vision-capable agent can attempt, and the rest of the workflow is deterministic code an agent can orchestrate. This moves automation in GRB analysis beyond a fixed heuristic computing a scalar score, in two ways: an *LLM agent makes the actual analysis decisions* rather than evaluating a pre-specified rule, and it does so across the *full pipeline*, so we can measure the downstream effect on the physics rather than agreement on a single label. The open question is not whether an agent *can* run the pipeline, but whether its judgements match an expert’s well enough that the science is unchanged—and whether the process can be made auditable rather than a black box.

A fair objection is that if these steps can be written down as task guides, they could simply be coded as a fixed-threshold pipeline, with no agent and no human. We take this possibility seriously, but it is precisely what the present practice already rejects: the reason GBM teams still place backgrounds and source intervals by eye is that a purely algorithmic version, committed to fixed thresholds, fails silently on a non-negligible fraction of bursts. In building the companion single-pulse analysis we ran exactly such a deterministic selector; on a sample of 106 bursts it placed the source window on pure background for several bursts—yielding spectral fits that are meaningless rather than merely imprecise—and produced over-wide background windows, which alias orbital trends into the polynomial baseline, for several dozen more. The failures are not random noise but structured judgement calls: distinguishing a clean detector from a contaminated one, an emission-free baseline from a soft tail, or a genuine onset from a noise-buried rise. The task guides specify *what to look for*, not a closed-form rule that resolves these calls on a specific noisy light curve—which is why they are criteria an expert applies rather than an algorithm. An agent is one candidate for supplying that residual judgement; the benchmark is agnostic to the mechanism, and a better deterministic method that matched the experts would be equally welcome. The contribution is not that an agent is irreplaceable, but that we *measure* whether one can stand in for the human at the gate—in selection agreement and in downstream physics—rather than assert it.

In this paper we (1) describe an agentic pipeline for single-pulse GBM GRBs built around a human-or-AI *approval gate* that makes every selection auditable; (2) use that gate to benchmark agent against expert on the

subjective steps and measure the downstream impact on spectral parameters; and (3) validate the pipeline by reproducing known single-pulse results. The single-pulse spectroscopy science itself is presented in a companion paper (Sharma et al., in prep.); here the focus is the *method* and its validation. Section 2 describes the sample and data; Section 3 the agentic pipeline; Section 4 the human–AI experiment; Section 5 the results; and Sections 6–7 the discussion and conclusions.

2. SAMPLE AND DATA

We analyze the single-pulse *Fermi*/GBM sample of the companion science paper: bursts selected by the shape-based “horizontal-line” criterion of Busby & Lazati (2024) (a fast-rise/exponential-decay single-pulse score), with a fluence cut to ensure adequate time-resolved signal-to-noise. The selection is on pulse *shape*, not brightness. We use the time-tagged event (TTE) data of the NaI detectors within $\sim 50^\circ$ of the source, the most-illuminated BGO, and, where available, LAT Low-Energy (LLE) data. For the human–AI benchmark (Section 4) we additionally define a subset of $N_{\text{bench}} = 25$ bursts processed in *both* approval modes. The subset is chosen deterministically (no random draws; the selection code and list are released with the repository) to stress the judgement steps rather than flatter them: six bursts on which a fixed-threshold selector demonstrably mis-placed the source window (Section 1), four reference bursts spanning the sample (including the brightest burst and the median-fluence case), and fifteen stratified fills — the faintest, median, and brightest burst of each sample fluence quintile. The result spans a factor of ~ 245 in fluence, durations of 7–453 s, and includes five bursts with LAT/LLE coverage.

3. THE AGENTIC PIPELINE

The pipeline is a self-documenting code repository that an AI coding agent reads and executes. A top-level machine-readable guide (an `AGENTS.md` file) orients any agent—human-invoked or autonomous—to the environment, data, run order, and the approval semantics, so a fresh agent with no prior context can drive the full chain. The analysis proceeds in three gated stages.

3.1. Stage 1: gated selection (detectors, background, source)

The selections that are traditionally manual are unified into one step that writes a single, stamped catalog. For each burst the agent (or a human) approves:

1. *Detectors*. Detector–source angles are computed from the spacecraft pointing (POSHIST quaternions); NaI detectors within a threshold angle plus the matching BGO are proposed and approved.

2. *Background.* Pre- and post-burst off-source windows are fit with a time polynomial (order chosen by a likelihood-ratio test) and interpolated beneath the burst, following standard GBM practice.
3. *Source.* The emission interval is marked explicitly per burst on the brightest-detector light curve.

Crucially, the approval is gated: every catalog row carries an *approval stamp* recording the approver and the mode—`human_gui` (a person marks the windows interactively) or `ai_vision` (the agent renders the light curves, inspects the images, and proposes the windows). An agent thus never silently substitutes its judgement for a human’s; the provenance of every decision is recoverable from the output file. This stamp is the foundation of the benchmark in Section 4.

3.2. Stage 2: adaptive binning

The light curve is partitioned with the Bayesian Blocks algorithm (Scargle et al. 2013) computed on the background-subtracted rate, combined with a per-block significance floor: blocks below the floor are merged so that every surviving spectral bin clears a detection threshold. The binning runs within the approved source interval.

3.3. Stage 3: spectral fitting and model selection

Each bin is fit with a family of empirical and photospheric models ([cite: Band 1993, CPL, SBPL/Ravasio, DSBPL, Band+BB, CPL+BB]) using a forward-folding likelihood with the detector responses and the polynomial background propagated as Gaussian errors (the `pgstat` statistic). The preferred model per bin is chosen by an information criterion with a physical-validity gate. [cite: 3ML / astromodels (Vianello et al.)]

3.4. Built-in adversarial verification

A distinguishing feature of the agentic implementation is that verification is itself automated: code changes are reviewed by an ensemble of independent agents that attempt to refute each other’s findings before a change is accepted. We report this protocol as part of the method (Appendix B), since it materially affects the reliability of an agent-built pipeline.

4. EXPERIMENTAL DESIGN: HUMAN VERSUS AI

The approval gate (Section 3.1) lets us run the subjective steps in both modes on the identical benchmark sample and hold everything downstream fixed. The human baseline is provided by two experienced GRB analysts (labelled Expert1 and Expert2), each approving the full benchmark subset independently, with no

discussion of selections until both catalogs are written. The inter-expert scatter sets the scale against which the agents are judged: an agent is consistent with the experts if it matches each of them about as well as they match each other. With two experts the inter-human reference is the distribution of the Expert1–Expert2 disagreement across the $N_{\text{bench}} = 25$ bursts; it is a per-burst distribution, not a single number, so an agent is compared against the spread, not a point. We define two families of metrics.

Systems benchmarked.—We evaluate several contemporary agentic coding systems—e.g. OpenAI Codex, Google Antigravity, and Anthropic Claude Code [PENDING run: fix exact versions]—each operating the same repository through the same machine-readable task guides (Section 3). Holding the instructions fixed across systems makes the comparison a test of the *models*, not of the prompts. Each system runs every judgement step (detector, background, source, QC) in `ai_vision` mode and writes its own stamped catalog; the human experts write theirs in `human_gui` mode. We then report, per metric, each system against the inter-expert band (a per-system leaderboard) and the systems against one another (do independent LLMs converge on the same selections?).

Selection agreement.—For each burst we compare each pair of raters: the approved-detector set agreement (Jaccard), the background-window overlap (intersection-over-union) and per-edge differences, and the source-interval edge differences and duration error. For every metric we report the AI-versus-human distribution against the human-versus-human distribution. [PENDING run: report values.]

Downstream impact (the decisive test).—We propagate each mode’s selections through the identical binning and fitting and compare the recovered quantities bin-by-bin and burst-by-burst: E_p , the low-energy index, the blackbody kT where a thermal component is preferred, and the thermal-versus-double-break classification. We quantify agreement with Pearson and Spearman statistics on the recovered parameters, referenced to the inter-human scatter. The question is whether scientific conclusions are invariant under the human→AI swap. [PENDING run: report the parameter scatter, correlation, and classification concordance.]

5. RESULTS

[PENDING run: This entire section awaits the dual-mode run. Planned content:]

- *Fig. 1*: per-burst selection-agreement summary (detector concordance; background- and source-edge differences).
- *Fig. 2*: AI-vs-human scatter of E_p , low-energy index, and kT with the one-to-one line and intrinsic scatter.
- *Table 1*: concordance of the per-bin model classification.
- *Fig. 3*: a representative burst shown end to end in both modes.
- *Validation*: recovery of established single-pulse relations (e.g. the E_p - kT correlation) under the agentic pipeline.

6. DISCUSSION

APPENDIX

A. REPRODUCIBILITY

The full pipeline, its agent guide (`AGENTS.md`), and the exact commands for each stage are released with this paper. The analysis can be re-run end to end by a human or an AI agent from the released repository. [\[cite: repository DOI / URL on publication.\]](#)

B. ADVERSARIAL VERIFICATION PROTOCOL

Code an agent writes must be checked by something other than the agent that wrote it. Throughout the construction of this pipeline we used a fixed multi-agent protocol for any change that could affect the science output, and we describe it here because it is part of what makes an agent-built pipeline trustworthy.

The protocol has three stages. (1) *Independent investigation*: several agents (typically three) examine the same question in parallel with deliberately different briefs—one reads the implementation end to end, one reasons about the general failure mode from first principles, one checks siblings and callers for the same defect—and each must ground every claim in file-and-line evidence. (2) *Synthesis*: a separate agent reconciles the investigators into a single root-cause statement and a minimal patch, re-reading the primary sources itself rather than trusting the summaries. (3) *Adversarial verification*: further agents are instructed to *refute* the patch—one attacks the root-cause claim (does the fix actually prevent the failure, traced concretely through the code paths?), one hunts for regressions (does any existing behaviour change?)—and are told to default to skepticism. A change is accepted only if the refuters fail.

Two design choices matter. The verifying agents receive different *lenses*, not identical instructions, because diverse perspectives catch failure modes that redundant ones miss; and refuters are rewarded for finding problems, inverting the confirmation bias that a single agent reviewing its own plan exhibits. As a concrete example from this project: a GUI crash reported by a collaborator (a Tcl/Tk error appearing only on Linux, only after the first of many windows) was diagnosed by this protocol to a cause none of the obvious hypotheses predicted—a module-level backend override in an imported prototype silently switching the GUI toolkit after startup—and the resulting two-line-plus-guard fix was confirmed by the refuting agents to preserve every interactive behaviour before it was deployed. The same protocol produced the audit findings that reshaped the companion paper’s statistical framework.

REFERENCES

- Busby, I., & Lazzati, D. 2024, *ApJ*, 972, 83
- Scargle, J. D., Norris, J. P., Jackson, B., & Chiang, J. 2013, *ApJ*, 764, 167

[PENDING run: Frame around the results.]

Anticipated axes: where the agent agrees with experts and where it does not (e.g. faint bursts, complex backgrounds, orbital trends); the reproducibility and auditability gains from the stamped gate; the throughput implications for scaling time-resolved catalogs; and an honest account of failure modes and the necessity of the human-in-the-loop option. Positioning relative to other agentic/LLM analyses in astronomy [\[cite: related work\]](#).

7. CONCLUSIONS

[PENDING run: Write after results.]

We present and benchmark an auditable agentic pipeline for single-pulse GBM time-resolved spectroscopy; the gated, human-or-AI design makes the traditionally manual choices reproducible and lets us measure, rather than assume, the impact of delegating them to an agent.